

Sergio Albacete

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Main research interests: pollinator ecotoxicology, environmental risk assessment, solitary bee biology, pollination ecology, climate change.

Summary:

I am an applied ecologist interested in understanding how global change drivers affect biodiversity, with a particular focus on solitary bees. My research integrates laboratory, semi-field and field experiments with ecological modelling to improve knowledge on pollinator ecology and support evidence-based environmental risk assessment (ERA).

I began my research career at IPE-CSIC, where during my bachelor's thesis I investigated how population size in threatened plant species influences the diversity and structure of their ecological interaction networks. During my Master's thesis at the University of Barcelona, I expanded this perspective by studying how human activities shape arthropod communities in riparian and forest ecosystems. In November 2024, I completed my PhD at CREAF under the joint supervision of Dr. Jordi Bosch and Dr. Anselm Rodrigo. My doctoral research focused on the effects of sublethal pesticide exposure on solitary bees, particularly *Osmia cornuta*, a model species proposed by EFSA for pesticide risk assessment in Europe. I contributed to the development of new methodologies for standardized toxicological testing in solitary bees, including an individual feeding method ("petal method") and a chronic-exposure protocol submitted by Spain to the OECD for potential regulatory adoption. I also co-led the design and validation of a phototaxis-based method to detect sublethal effects, already applied in several studies. Under semi-field and field conditions, I studied how pesticide mixtures influence nesting behaviour, reproductive success and population dynamics. I additionally explored interactions between pesticides and climate change. Since January 2025, I have been conducting my postdoctoral research at the University of Bologna (Italy) on projects focused on improving ERA for wild pollinators, including wild bees, hoverflies and butterflies.

EDUCATION	
2020 – 2024	**PhD in Terrestrial Ecology at CREAF (Centre for Research on Ecology and Forestry Application). (Excellent *Cum Laude*).** "Solitary Bees in a Changing World: Sublethal effects of pesticide exposure and climate change on the solitary bee *Osmia cornuta*" Autonomous University of Barcelona (UAB).
2017 – 2018	**Master's degree in Biodiversity and Conservation.** Specialization: Conservation biology and biodiversity management. University of Barcelona (UB)

	Master's thesis: "The importance of landscape heterogeneity and local conditions in Galician chestnut plantations and their conservation value for arthropod communities" Albacete et al., 2020 FEM
2013 – 2017	**Bachelor's degree in Environmental Science.** Specialization: Management of the natural environment. University of Barcelona (UB)

PROFESSIONAL AND SCIENTIFIC EXPERIENCE	
2025 – Present	**Postdoctoral researcher** at Department of Agricultural and Food Science, University of Bologna (UNIBO, Italy). Collaboration with Dr. Fabio Sgolastra.
2024 (Mar. – June)	**PhD research internship** at Evolution, Genome, Behavior and Ecology laboratory (EGCE-IDEEV), Paris-Saclay University (France). Collaboration with Dr. Fabrice Requier. Modeling climate change and pesticide effects on solitary bee populations using SolBeePop.
2020 – 2024	**Predoctoral researcher at** CREAF (Centre for Research on Ecology and Forestry Application), Autonomous University of Barcelona (UAB, Spain). Supervised by Dr. Jordi Bosch & Dr. Anselm Rodrigo.
2020 (Jan. – June)	**Technician** in the Insect and Pollination Ecology group at Centre for Research on Ecology and Forestry Application (CREAF), Autonomous University of Barcelona (UAB, Spain).
2019 (Feb. – Dec.)	**Environmental consultant** at EDAFO GM, S.A.
2018 (Jan. – Oct.)	**Master's research internship** at the Evolutionary Biology, Ecology, and Environmental Sciences department (BEECA), University of Barcelona (UB, Spain). Supervised by Dr. Alberto Maceda-Veiga.
2017 (Apr. – Sep.)	**Bachelor's research internship** at the Pyrenean Institute of Ecology (IPE-CSIC, Spain). Supervised by Dr. Mª Begoña García.

RESEARCH PROJECTS	
2026 – 2029	**Solitary bees under climate change.** Spanish Ministry of Ministry of Science, Innovation and Universities, R+D National Program. Project PID2025-174249NB-I00. (Participation as Postdoctoral researcher).
2026 – 2027	**Beyond bloom: Impacts of authorized insecticide applications during crop petal fall on wild pollinator communities**. IUCN Funding - European Fund for Youth Action on Pollinators. Amount: 15,000 €. (Participation as Project leader).
2024 – 2027	**PollinERA. Understanding pesticide-Pollinator interactions to support EU Environmental Risk Assessment and policy.** Project HORIZON-CL6-2023-BIODIV-01-1 Area B. (Participation as Postdoctoral researcher).
2024 – 2026	**Estimates of food consumption in bees.** European Food Safety Authority. Project EPA13.02-PREV-11. (Participation as Project officer).

2022 – 2025	**Combined effects of pesticide exposure and other stress factors on solitary bees.** Spanish Ministry of Science and Innovation, R+D National Program. Project PID2021-128938OB-I00. (Participation as PhD student).
2019 – 2021	**Behavioral and populational effects of fungicide sprays on fruit tree pollinators: laboratory, cage, and field studies.** Spanish Ministry of Science and Innovation, R+D National Program. Project I+D+I. RTI2018-098399-B-I00 (Participation as PhD student).
2015 – 2018	**PERDIVER. Losing the network. The role of threatened plants in the conservation of global diversity.** BBVA Foundation, Pyrenean Institute of Ecology-CSIC. (Participation as Internship student).

PUBLICATIONS

Publications peer reviewed – SCI:

- Albacete, S., Azpiazu, C., Sancho, G., García-Gómez, A., Sgolastra, F., Rodrigo, A., & Bosch, J. (2026). From the spray tank to the bee nest: translocation and persistence of fungicide residues in *Osmia cornuta* food provisions. *Environmental Pollution*, 129357. https://doi.org/10.1016/j.envpol.2026.129357

- Topping, C., Focks, A., Albacete, S., Segovia, C. A., Bednarska, A., Benrezkallah, J., ... & Axelman, J. (2026). Trait-based specific protection goals for wild pollinators: a staged modelling framework for pesticide risk assessment. Research Square (Under revision, *Environmental Sciences Europe*). https://doi.org/10.21203/rs.3.rs-10019161/v1

- Parkinson, R. H., Cerbone, H., Mieskolainen, M., Cao, S., Wilson, A. D., Albacete, S., ... & Roberts, S., 2026. MetaBeeAI: An AI pipeline for structured evidence extraction from biological literature. *Ecological Informatics*, 103813. https://doi.org/10.1016/j.ecoinf.2026.103813

- Miranda-Cebrián, H., Cáliz, J., Olesen, J. M., Megias, A. G., Albacete, S., Martín, F., ... & García, M. B., 2026. Rare plants sustain communities of visitors with similar diversity but different composition across populations of different size. *Perspectives in Plant Ecology, Evolution and Systematics*, 125930. https://doi.org/10.1016/j.ppees.2026.125930

- Sancho, G., Albacete, S., Azpiazu, C., Sgolastra, F., Rodrigo, A., & Bosch, J., 2025. A phototaxis assay to measure sublethal effects of pesticides on bees. *Scientific Reports*, 15(1), 22136. https://doi.org/10.1038/s41598-025-05400-7.

- Tadei, R., Cilia, G., Mathias da Silva, E. C., Sancho, G., Albacete, S., Azpiazu, C., ... & Sgolastra, F., 2025. Co-exposure to a honeybee pathogen and an insecticide: synergistic effects in a new solitary bee host but not in *Apis mellifera*. *Proceedings of the Royal Society B*, 292(2042), 20242809. https://doi.org/10.1098/rspb.2024.2809

- Albacete, S., Azpiazu, C., Sancho, G., Barnadas, M., Alins, G., Sgolastra, F., Rodrigo, A. & Bosch, J., 2024. Sublethal fungicide-insecticide co-exposure affects nest recognition and parental investment in a solitary bee. *Environmental Pollution*, 125223. https://doi.org/10.1016/j.envpol.2024.125223

- Albacete, S., Sancho, G., Azpiazu, C., Rodrigo, A., Sgolastra, F., & Bosch, J., 2024. Exposure to sublethal levels of insecticide-fungicide mixtures affect reproductive success and population growth rates in the solitary bee *Osmia cornuta*. *Environment International*, 190, 108919. https://doi.org/10.1016/j.envint.2024.108919

- Azpiazu, C., Sgolastra, F., Ippolito, A., Albacete, S., Brandt, A., Colli, M., Grossar, D., Jeker, L., Malagnini, V., Sancho, G., Splitt, A., Straub, L., Strobl, V., Boranski, M., Jachuła, J., Martins, C., Medrzycki, P., Simon-Delso, N., Tosi, S. & Bosch, J., 2024. Chronic oral toxicity protocol for adult solitary bees (*Osmia bicornis* L.): reduced survival under long-term exposure to a "bee-safe" insecticide. *Environmental Pollution*, 125129. https://doi.org/10.1016/j.envpol.2024.125129

- Albacete, S., Sancho, G., & Bosch, J. (2024) Use of a novel nesting material by the spider wasp *Dipogon variegatus* (Hymenoptera, Pompilidae). *Journal of Hymenoptera Research,* 97, 541-544. https://doi.org/10.3897/jhr.97.123853

- Albacete, S., Sancho, G., Azpiazu, C., Rodrigo, A., Molowny-Horas, R., Sgolastra, F., & Bosch, J. (2023). Bees exposed to climate change are more sensitive to pesticides. *Global change biology*, 29(22), 6248-6260. https://doi.org/10.1111/gcb.16928 (Featured in: *Nature Climate Change*)

- Azpiazu, C., Hinarejos, S., Sancho, G., Albacete, S., Sgolastra, F., Martins, C. A. H., ... & Bosch, J. (2023). Description and validation of an improved method to feed solitary bees (*Osmia spp.*) known amounts of pesticides. *Ecotoxicology and Environmental Safety*, 264, 115398. https://doi.org/10.1016/j.ecoenv.2023.115398

- Maceda-Veiga, A., Albacete, S., Flor-Arnau, N., Vieira, C., Bros, V., Domènech, M., ... & Mac Nally, R. (2021). Local and downstream cumulative effects of traditional meadow management on stream-water quality and multiple riparian taxa. *Science of The Total Environment*, 794, 148601. https://doi.org/10.1016/j.scitotenv.2021.148601

- Maceda-Veiga, A., Albacete, S., Carles-Tolrá, M., Pujade-Villar, J., Máca, J., Mac Nally, R. (2021). Streams and rural abandonment are related to the summer activity of the invasive pest *Drosophila suzukii* in protected European forests. *Forest Ecology and Management*, 485, 118942. https://doi.org/10.1016/j.foreco.2021.118942

- Matas, A., Mac Nally, R., Albacete, S., Carles-Tolrá, M., Domènech, M., Vives, E., ... & Maceda-Veiga, A. (2021). Wild boar rooting and rural abandonment may alter food-chain length in arthropod assemblages in a European forest region. *Forest Ecology and Management*, 479, 118583. https://doi.org/10.1016/j.foreco.2020.118583

- Albacete, S., Mac Nally, R., Carles-Tolra, M., Domenech, M., Vives, E., Espadaler, X., ... & Maceda-Veiga, A. (2020). Stream distance and vegetation structure are among the major factors affecting various groups of arthropods in non-riparian chestnut forests. *Forest Ecology and Management*, 460, 117860. https://doi.org/10.1016/j.foreco.2019.117860

Book chapters:

- García, M. B., Miranda, H., Jarne, M., Albacete, S., del Arco, L., Martín, F., Múgica, A., González, A. & Olesen, J. (2024). "Interacciones biológicas en ambientes raros: plantas rupícolas." In Más allá de la pérdida de especies: Interacciones ecológicas en el antropoceno. Medel, R., Traveset, A., Navarro, L. (editors). Ediciones Fremen SpA., Santiago de Chile. pp. 316-334.

Publication peer reviewed:

- Carles-Tolrá, M., Albacete, S., Matas, A., Maceda-Veiga, A. (2019) *Piezura pardalina* Rondani, 1866 y *Mesembrina intermedia* Zetterstedt, 1849, dos especies nuevas para España, eliminación de *Sapromyza nitida* Czerny, 1932 de la lista española (Diptera: Lauxaniidae, Fanniidae, Muscidae). *Arquivos Entomolóxicos*, (21), 217-219.

- Carles-Tolrá, M., Albacete, S., Jarne, M. (2019) *Sargus rufipes* Wahlberg, 1854: una especie nueva para la Península Ibérica (Diptera: Stratiomyidae). *Arquivos Entomolóxicos*, (21), 3-4.

- Carles-Tolrá, M., Albacete, S., Maceda-Veiga, A. (2018) Tres especies nuevas de dípteros para España (Diptera: Lauxaniidae, Periscelididae, Fanniidae). *Arquivos Entomolóxicos*, (19), 167-168.

CONFERENCES

- Hernández-Márquez, A., del Arco, L., Usuelli, L., Sgolastra, F., Rodrigo, A., Bosch, J., Albacete, S. Beyond bloom: impacts of authorized insecticide applications during crop petal fall on wild pollinator communities. **EurBee 11**. Bologna, Italy, September 2026. Poster.

- Tadei, R., Albacete, S., Cabiddu, L., Girodo, R., Trejo, S., Di Noi, A., Caliani, I., Giovanetti, L., Campani, T., Casini, S., Bosch, J., Sgolastra, F. Comparing lethal and sublethal endpoints in two bee species exposed to flupyradifurone. **EurBee 11**. Bologna, Italy, September 2026. Talk.

- García-Gómez, A., Ciot, L., Albacete, S., Barnadas, M., Rodrigo, A., Bosch, J. Pollen composition of larval food provisions affects development and adult insecticide sensitivity in a polylectic solitary bee. **EurBee 11**. Bologna, Italy, September 2026. Talk.

- Albacete, S., Sancho, G., Azpiazu, C., Rodrigo, A., Sgolastra, F., & Bosch, J. Exposure to realistic levels of pesticide mixtures affect nesting performance, longevity, and reproductive success in a solitary bee. **XII European Congress of Entomology**. Heraklion, Crete, Greece, October 2023. Talk.

- Sancho, G., Albacete, S., Azpiazu, C., Sgolastra, F., Rodrigo, A., & Bosch, J. Pesticide loads in food provisions of the solitary bee *Osmia cornuta*: effects of landscape composition and management practices. **XII European Congress of Entomology**. Heraklion, Crete, Greece, October 2023. Talk.

- Albacete, S., Sancho, G., Azpiazu, C., Sgolastra, F., Rodrigo, A. & Bosch, J. Bees exposed to climate change are more sensitive to pesticides. **XII National Congress of Applied Entomology (SEEA)**. Málaga, Spain, October 2022. Talk.

- Sancho, G., Albacete, S., Azpiazu, C., Sgolastra, F., Rodrigo, A. & Bosch, J. A phototaxis method to measure sublethal effects of pesticides on bees. **XII National Congress of Applied Entomology (SEEA)**. Málaga, Spain, October 2022. Talk.

- Azpiazu, C., Hinarejos, S., Sancho, G., Albacete, S., Sgolastra, F., Martins, C.A.H., Domene, X., Rodrigo, A. & Bosch, J. Description, and validation of an improved method to feed solitary bees (*Osmia spp.*) known amounts of pesticides. **XII National Congress of Applied Entomology (SEEA)**. Málaga, Spain, October 2022. Talk.

THESES SUPERVISED

PhD thesis:

- A. Hernández-Márquez. Autonomous University of Barcelona (UAB). **PhD thesis**. Supervisors: Bosch, J., Albacete, S. (In progress, 2025-2029)

Master thesis:

- L. Usuelli. Università di Milano. **Master thesis**. Supervisors: Bosch, J., Albacete, S., Biella, P. (In progress, 2026-2027)

- M. Mislej. Università di Bologna. **Master thesis**. Supervisors: Sgolastra, F., Albacete, S., Cabiddu, L. (In progress, 2026-2027)

- D. J. Pamulaklakin. Università di Bologna. **Master thesis**. Supervisors: Sgolastra, F., Albacete, S, Tadei, R., Girodo, R. (October 2026)

- L. Signoroni. Università di Milano. **Master thesis** Supervisors: Bosch, J., Albacete, S., Biella, P. (November 2026).

- A. Cavallini. Università di Bologna. **Master thesis**. Supervisors: Sgolastra, F., del Arco, L., Cabiddu, L., Albacete, S. (December 2025).

Bachelor thesis:

- F. Mastroberti. Università di Bologna. **Bachelor thesis**. Supervisors: Sgolastra, F., Cabiddu, L., Albacete, S., del Arco, L. (July 2025).